

CLINICAL INTELLIGENCE TREND REPORT

RED: Provider Review
Recommended

Patient ID: 934297 0122 Period: June 24 to June 30, 2024 Report Date: July 1, 2024 Coverage: 122/168h (72.6%)

Decision support summary derived from longitudinal vital sign trends; interpret in clinical context.

SECTION 1 — Clinical Intelligence: Signal Summary

During the reporting period (June 24 to June 30, 2024), 122 of 168 expected hours were recorded (72.6% coverage). 9 episodic events spanning 38 total hours were detected, including: Elevated Breathing Rate, High Heart Rate, Low Heart Rate, and Very Low Heart Rate. Events occurred in clustered bursts, with the longest persisting 10 consecutive hours. Heart rate ranged from 47 to 101 bpm (P5 to P95), a 55 bpm spread spanning nearly the full adult range within a single week. 4 episodes showed concurrent low heart rate with elevated breathing rate, a coupling pattern that may reflect compensatory respiratory stress and warrants closer clinical evaluation.

Trend Assessment: *Progressively unstable vital sign pattern* Action Posture: *Provider review advised*

Suggested Clinical Review Actions

- June 24 to 25: sustained low heart rate below 45 bpm for 4+ hours with concurrent elevated breathing rate. Review heart rate lowering medications, check blood pressure, assess oxygen levels, lung sounds, and fluid status.
- June 27 evening: abrupt shift to high heart rate above 100 bpm for 7 hours. Evaluate for pain, infection, fever, dehydration, or new heart rhythm change.
- Overall: heart rate spanning 55 bpm in one week (P5=47, P95=101) with escalating episode frequency suggests progressive instability; consider earlier provider follow up.

SECTION 2 — High Priority Episodic Events

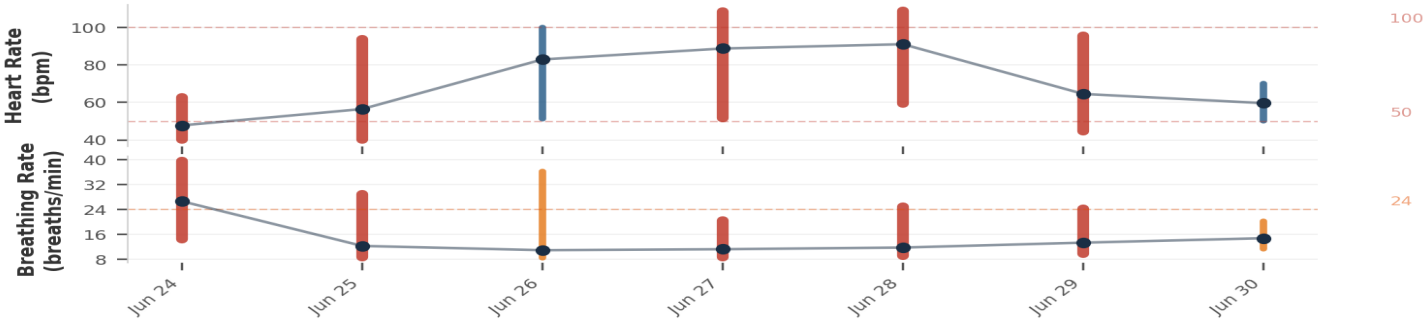
Event Type	Date / Time Window	Dur	Key Vitals	Clinical Concern
Very Low Heart Rate	06/24 07:00 to 06/24 10:00	4h	HR 48 Min 42 RR 38	Sustained pattern; consider provider review (HR+RR coupled)
Elevated Breathing Rate	06/24 00:00 to 06/24 09:00	10h	HR 47 Min 40 RR 40	Sustained deviation; review context (HR+RR coupled)
Low Heart Rate	06/24 07:00 to 06/24 11:00	5h	HR 48 Min 42 RR 38	Sustained deviation; review context (HR+RR coupled)
Very Low Heart Rate	06/25 01:00 to 06/25 05:00	5h	HR 45 Min 40 RR 29	Sustained deviation; review context
High Heart Rate	06/27 20:00 to 06/28 02:00	7h	HR 102 Min 91 RR 19	Sustained deviation; review context
Very Low Heart Rate	06/29 06:00 to 06/29 06:00	1h	HR 50 Min 44 RR 22	Sustained deviation; review context

SECTION 3 — Heart Rate and Breathing Rate Statistics

Metric	Mean	Min	Max	P5	P95
Heart Rate Avg (bpm)	74.5	41.3	103.6	46.6	101.4
Heart Rate Min (bpm)	65.4	40.1	99.5	40.7	93.8
Breathing Rate Avg (brpm)	13.5	9.7	29.1	10.1	28.2
Breathing Rate Max (brpm)	19.3	11.2	39.6	12.4	38.2

SECTION 4 — Visual Trend Analysis

Daily Vital Sign Ranges (Red bars = episode days; dots = daily average; dashed lines = clinical thresholds)



Hourly Average Distribution (Red arrow shows the P5 to P95 range; a wide spread indicates high cardiac variability)

